General Mathematics /Numerical Ability

43. How many integers are there between 1 and 100 which have 4 as a digit but are not divisible by 4?
(a) 5 (b) 11
(c) 12 (d) 13

Ans. (c) : Such integers, there between 1 and 100 which have 4 as a digit but are not divisible by 4:-
14, 34, 41, 42, 43, 45, 46, 47, 49, 54, 74, 94
∴ The number of such integers = 12

44. Let x, y be the volumes m, n be the masses of two metallic cubes P and Q respectively. Each side of Q is two times that of P and mass of Q is two times that of P. Let u = m/x and v = n/y. Which one of the following is correct?
(a) u = 4v (b) u = 2v
(c) u = v (d) v = 4u

Ans. (a) : ∵ Mass of P = m, Mass of Q = n
Volume of P = x, Volume of Q = y
∴ Density of P = $\frac{m}{x}$, Density of Q = $\frac{n}{y}$
and $\boxed{u=\frac{m}{x}}$, $\boxed{v=\frac{n}{y}}$
Let side of P = a
∴ Side of Q = 2a
Volume of P,
$X = a^3$
and volume of Q,
$y = 8a^3$
According to question, n = 2m
$\boxed{n=2m}$
$u=\frac{m}{x}$ $v=\frac{n}{y}$
$\boxed{u=\frac{m}{a^3}}$ $\boxed{v=\frac{2m}{8a^3}}$
$u:v=\frac{m}{a^3}:\frac{2m}{8a^3}$
u : v = 4 : 1
$\boxed{u=4v}$

45. The average age of a teacher and three students is 20 years. If all the three students are of same age and the difference between the age of the teacher and each student is 20 years, then what is the age of the teacher?
(a) 25 years (b) 30 years
(c) 35 years (d) 45 years

Ans. (c) : Let the age of teacher = x years then the age of each student = (x – 20)
According to question,
$$\frac{(x-20)+(x-20)+(x-20)+x}{4}=20$$
4x – 60 = 80
4x = 140
$\boxed{x=35}$
Hence the age of the teacher = 35 years.

46. A person bought a car and sold it for ₹ 3,00,000. If he incurred a loss of 20%, then how much did he spend a buy the car?
(a) ₹ 3,00,000 (b) ₹ 3,65,000
(c) ₹ 3,70,000 (d) ₹ 3,75,000

Ans. (d) : ∵ SP = 300000 ₹
and loss % = 20%
$$\because CP = SP\times\frac{100}{(100-L)}$$
$$CP = 30,0000\times\frac{100}{80}$$
$$= 300000\times\frac{5}{4}$$
= 75000 × 5
CP = 375000 ₹

47. Two Statement S_1 and S_2 are given below with regard to two numbers followed by a Question :
S_1 : Their product is 21
S_2 : Their sum is 10
Question :
What are the two numbers?
Which one of the following is correct in respect of the above Statements and the Question?
(a) S_1 alone is sufficient to answer the Question.
(b) S_2 alone is sufficient to answer the Question.
(c) S_1 and S_2 together are sufficient to answer the Question, but neither S_1 alone nor S_2 alone is sufficient to answer the Question.
(d) S_1 and S_2 together are not sufficient to answer the Question.

Ans. (c) : Let the numbers a, b
From statement (S_1):-
a × b = 21 _____(i)
From statement (S_2):-
a + b = 10_____(ii)
∵ $(a-b)^2=(a+b)^2-4ab$
$(a-b)^2=(10)^2-4\times21$
$(a-b)^2=100-84$
$(a-b)^2=16$
a – b = 4_____(iii)
from eq (ii) & (iii)
$\boxed{a=7}$ and $\boxed{b=3}$
Hence,
S_1 and S_2 together are sufficient to answer the question, but neither S_1 alone nor S_2 alone is sufficient to answer the question.

48. If you have to straight sticks of length 7.5 feet and 3.25 feet, what is the minimum length can you measure?
(a) 0.05 foot (b) 0.25 foot
(c) 1 foot (d) 3·25 feet

Ans. (b) : For minimum length,
taken H.C.F. 7.5 feet and 3.25 feet.
On equalizing the denominator-

$$\Rightarrow 7.5=\frac{750}{100}$$
$$\Rightarrow 3.25=\frac{325}{100}$$
$$\text{H.C.F.}=\frac{(750,325)\text{H.C.F.}}{(100,100)\text{L.C.M.}}$$
$$\text{H.C.F.}=\frac{25}{100}$$
H.C.F. = 0.25 foot
Hence, the minimum length you can measure = 0.25 foot.

49. A simple mathematical operation in each number of the sequence 14, 18, 20, 24, 30, 32, ... results in a sequence with respect to prime numbers. Which one of the following is the next number in the sequence?
(a) 34 (b) 36
(c) 38 (d) 40

Ans. (c) :

Sequence :–	14,	18,	20,	24,	30,	32	38
	↑+1	↑+1	↑+1	↑+1	↑+1	↑+1	↑+1
Prime number :–	13	17	19	23	29	31	37

50. One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers o the remaining pages is 195. The torn page contains which of the following numbers?
(a) 5, 6 (b) 7, 8
(c) 9, 10 (d) 11, 12

Ans. (b) : Let there are n pages in a book.
According to question,
1 + 2 + 3 + 4 + 5……n pages= 195 + two pages
$$\frac{n(n+1)}{2}=195+\text{the sum of two pages}$$
$$\because \text{The sum of first n natural numbers}=\frac{n(n+1)}{2}$$
Put n = 20
Then
$$\frac{20(20+1)}{2}=195+\text{the sum of two pages.}$$
10 × 21 = 195 + the sum of two pages
The sum of two pages = 210 – 195
= 15
∵ The sum of the number of torn pages = 15
∴ The page number of torn pages = 7, 8

51. Consider the following data:

	Average marks in English	Average marks in Hindi
Girls	**9**	**8**
Boys	**8**	**7**
Overall average marks	**8.8**	**x**

What is the value of x in the above table?
(a) 7·8 (b) 7·6
(c) 7·4 (d) 7·2

Ans. (a) : Let the number of girls = g
and the number of boys = b
According to question,
$$\text{Average}=\frac{9\times g+8\times b}{(g+b)}$$
$$(g+b)(8.8)=9g+8b$$
$$8.8g+8.8b=9g+8b$$
$$8.8b-8b=9g-8.8g$$
$$0.8b=0.2g$$
$$8b=2g$$
$$4b=g$$
b : g = 1 : 4
Again,
Overall average marks in Hindi
$$x=\frac{8\times g+7\times b}{g+b}$$
$$x=\frac{8\times4+7\times1}{1+4}$$
$$x=\frac{32+7}{5}$$
$$x=\frac{39}{5}$$
$\boxed{x=7.8}$

52. How many zeroes are there at the end of the following product?
1 × 5 × 10 × 15 × 20 × 25 × 30 × 35 × 40 × 45 × 50 × 55 × 60
(a) 10 (b) 12
(c) 14 (d) 15

Ans. (a) : ∵ Multiplying 2nd 5 gives zero.
∴ Find the power of 2 and 5, In 2 and 5 whose power is less, the same number of zeros will be obtain.
⇒1×5×10×15×20×25×30×35×40×45×50×55×60
Power of 5 = 5^{14}
and Power of 2 = 2^{10}
∵ The number of zeros = 10
Note:-
Power of 5
$5\to1$, $10\to1$, $15\to1$, $20\to1$, $25\to2$, $30\to1$, $35\to1$, $40\to1$, $45\to1$, $50\to2$, $55\to1$, $60\to1$
$\Rightarrow 5^{14}$
Power of 2
$10\to1$, $20\to2$, $30\to1$, $40\to3$, $50\to1$, $60\to2$
2^{10}